

UPERNAVIK, Greenland

WMO: 042110

Lat: 72.783N

Lon: 56.133W

Elev: 129

StdP: 99.78

Time Zone: -3.00 (GRL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	-30.4	-28.6	-33.0	0.2	-30.3	-31.2	0.2	-28.4	19.4	-6.0	14.1	-11.6	2.3	60	0.956

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.0	14.0	7.7	12.6	7.1	11.3	6.5	8.4	12.4	7.7	11.2	7.1	10.3	2.9	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.7	6.2	8.0	5.9	5.8	7.5	5.1	5.5	7.2	25.9	12.5	24.4	11.4	23.2	10.5	12.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.3	12.7	10.1	DB	-29.2	16.2	2.5	1.8	-31.0	17.5	-32.4	18.5	-33.8	19.5	-35.6	20.8	(4)
(5)			WB	-29.4	10.0	2.5	1.4	-31.2	11.0	-32.7	11.8	-34.1	12.6	-36.0	13.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-5.6	-15.6	-17.9	-18.3	-11.3	-3.1	3.7	7.2	6.5	1.9	-3.2	-7.0	-10.8		
	DBStd	9.98	6.31	7.10	6.58	5.29	4.14	3.63	3.16	2.55	2.49	3.18	3.69	4.78		
	HDD10.0	5708	792	782	877	640	406	191	99	113	243	409	510	646		
	HDD18.3	8733	1051	1015	1136	890	664	439	347	367	493	668	760	904		
	CDD10.0	17	0	0	0	0	0	2	11	4	0	0	0	0		
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	4.0	4.2	3.4	3.4	3.9	3.6	3.6	3.3	3.5	3.9	4.9	5.5	5.2		
	PrecAvg	242	11	11	8	10	10	15	25	30	36	31	32	20		
(15) Precipitation	PrecMax	351	53	40	18	29	36	47	74	96	101	70	114	44		
	PrecMin	121	0	0	0	0	1	1	0	1	2	3	1	0		
	PrecStd	65	11	11	6	8	8	13	20	23	28	17	25	12		
	PrecStd	65	11	11	6	8	8	13	20	23	28	17	25	12		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.7	2.4	2.8	4.9	10.1	14.6	16.6	14.9	8.9	5.9	5.0	2.2		
		MCWB	-0.6	-0.9	0.3	2.0	5.2	7.8	9.0	8.4	4.2	2.1	2.6	-0.4		
	2%	DB	-1.2	-1.2	-1.1	0.9	7.0	12.6	14.6	13.1	7.1	4.0	1.7	-0.2		
		MCWB	-3.0	-3.3	-3.3	-1.4	2.9	6.9	8.2	7.2	3.1	1.1	-0.2	-2.3		
	5%	DB	-4.2	-4.3	-5.1	-1.7	4.8	11.0	13.2	11.8	5.9	2.5	0.0	-2.3		
		MCWB	-6.0	-5.6	-6.8	-3.5	1.9	5.9	7.5	6.8	3.0	0.4	-1.8	-3.6		
	10%	DB	-7.2	-7.6	-9.4	-4.2	2.3	9.4	11.9	10.3	5.0	0.9	-1.5	-4.7		
		MCWB	-8.4	-8.6	-10.2	-5.3	0.5	4.9	6.8	6.2	2.7	-0.4	-2.8	-5.9		
	0.4%	WB	0.2	-1.0	0.2	2.1	5.6	8.2	9.7	9.4	5.1	2.8	3.2	0.3		
		MCDB	0.8	1.7	1.4	4.6	9.3	13.3	14.7	13.0	6.9	4.7	4.7	2.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-2.7	-3.4	-4.0	-1.6	3.3	7.2	8.7	8.1	4.4	1.3	0.2	-1.9		
		MCDB	-1.3	-1.6	-2.3	0.5	6.0	12.0	13.0	11.1	6.0	3.0	1.4	-0.6		
	5%	WB	-5.6	-6.5	-7.6	-3.4	2.0	6.2	8.1	7.5	3.8	0.4	-1.3	-3.5		
		MCDB	-4.0	-5.3	-6.3	-1.7	4.5	10.4	11.9	10.2	5.2	1.9	-0.2	-2.0		
	10%	WB	-8.1	-9.0	-10.5	-5.4	0.6	5.2	7.4	6.9	3.0	-0.5	-2.8	-5.6		
		MCDB	-7.0	-8.0	-9.8	-4.2	2.2	8.6	10.9	9.4	4.5	0.7	-1.4	-4.5		
	5% DB	MDBR	3.8	4.5	4.7	4.9	4.0	4.9	5.0	3.9	2.8	2.5	2.8	3.0		
		MCDBR	7.1	8.1	7.4	7.5	7.4	7.3	6.9	6.2	3.9	4.3	4.9	5.3		
		MCWBR	5.9	6.1	5.9	5.7	4.4	3.7	3.4	3.3	2.6	3.1	4.0	4.2		
		MCWBR	7.1	7.5	6.9	7.5	6.9	7.3	6.3	5.4	3.4	3.6	5.2	5.4		
(35) Mean Daily Temperature Range	5% WB	MCWBR	6.0	6.3	6.0	6.0	4.4	4.0	3.5	3.2	2.6	3.0	4.5	4.4		
		taub	N/A	0.158	0.236	0.282	0.324	0.324	0.316	0.309	0.278	0.356	N/A	N/A		
		taud	N/A	1.941	2.160	2.126	2.196	2.455	2.550	2.555	2.489	2.938	N/A	N/A		
		Ebn at Noon	0	479	739	822	834	858	855	812	714	409	0	0		
(42) Clear-Sky Solar Irradiance	Edh at Noon		0	27	64	100	109	89	78	67	50	23	0	0		
			0	27	64	100	109	89	78	67	50	23	0	0		
(44) All-Sky Solar Radiation	RadAvg	0.00	0.21	1.69	4.22	6.06	6.48	5.54	3.42	1.65	0.38	0.00	0.00	0.00		
	RadStd	0.00	0.02	0.08	0.26	0.31	0.51	0.49	0.38	0.11	0.03	0.00	0.00	0.00		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page